MAD Practical Solutions

Mobile Application Development – Android Studio (Java)

Q1. Linear Layout & Absolute Layout

Paste XML in: res/layout/activity_main.xml | Paste Java in: MainActivity.java

Part A – Linear Layout (Vertical)

File: res/layout/activity_linear.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: John Doe"
        android:textSize="18sp"
        android:padding="8dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Age: 21"
        android:textSize="18sp"
        android:padding="8dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Mobile: 9876543210"
        android:textSize="18sp"
        android:padding="8dp"/>

</LinearLayout>
```

Part B – Absolute Layout (Centred)

File: res/layout/activity_absolute.xml

AbsoluteLayout is deprecated but still works for lab purposes.

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: John Doe"
        android:textSize="18sp"
        android:layout x="100dp"
        android:layout_y="150dp"/>

    <TextView
        android:layout width="wrap content"
        android:layout_height="wrap_content"
        android:text="Age: 21"
        android:textSize="18sp"
        android:layout x="130dp"
        android:layout_y="200dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Mobile: 9876543210"
        android:textSize="18sp"
        android:layout x="80dp"
        android:layout_y="250dp"/>

</AbsoluteLayout>
```

File: MainActivity.java (same for both – just set the layout)

```
package com.example.layouts;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        // For Linear Layout:
        setContentView(R.layout.activity_linear);
        // For Absolute Layout, change to:
        // setContentView(R.layout.activity_absolute);
    }
}
```

Q2. Table Layout – 10 Students Basic Information

File locations: res/layout/activity_main.xml and MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TableLayout
        android:id="@+id/tableLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:stretchColumns="*"
        android:padding="8dp">

        <!-- Header Row -->
        <TableRow android:background="#1565C0" android:padding="4dp">
            <TextView android:text="No." android:textColor="#FFFFFF"
android:textStyle="bold" android:padding="4dp"/>
            <TextView android:text="Name" android:textColor="#FFFFFF"
android:textStyle="bold" android:padding="4dp"/>
            <TextView android:text="Roll No" android:textColor="#FFFFFF"
android:textStyle="bold" android:padding="4dp"/>
            <TextView android:text="Marks" android:textColor="#FFFFFF"
android:textStyle="bold" android:padding="4dp"/>
        </TableRow>

    </TableLayout>
</ScrollView>
```

File: MainActivity.java

```
package com.example.tablelayout;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
import android.widget.TableLayout;
import android.widget.TableRow;
import android.widget.TextView;
import android.graphics.Color;

public class MainActivity extends AppCompatActivity {

    String[] names  = {"Alice","Bob","Charlie","Diana","Eve",
                       "Frank","Grace","Henry","Ivy","Jack"};
    String[] rolls  = {"101","102","103","104","105","106","107","108","109","110"};
    String[] marks  = {"85","90","78","92","88","76","95","83","70","89"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        TableLayout table = findViewById(R.id.tableLayout);

        for (int i = 0; i < names.length; i++) {
            TableRow row = new TableRow(this);
            row.setBackgroundColor(i % 2 == 0 ? Color.parseColor("#E3F2FD") :
Color.WHITE);

            addCell(row, String.valueOf(i + 1));
            addCell(row, names[i]);
            addCell(row, rolls[i]);
            addCell(row, marks[i]);

            table.addView(row);
        }
    }

    private void addCell(TableRow row, String text) {
        TextView tv = new TextView(this);
        tv.setText(text);
        tv.setPadding(8, 8, 8, 8);
        tv.setTextSize(14);
        row.addView(tv);
    }
}
```

Q3. Accept Username and Password using TextView and EditText

Paste XML in res/layout/activity_main.xml and Java in MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout width="match parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Username:"
        android:textSize="16sp" android:textStyle="bold"/>

    <EditText
        android:id="@+id/etUsername"
        android:layout width="match parent"
        android:layout_height="wrap_content"
        android:hint="Enter username"
        android:inputType="text"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Password:"
        android:textSize="16sp" android:textStyle="bold"
        android:layout_marginTop="12dp"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout width="match parent"
        android:layout_height="wrap_content"
        android:hint="Enter password"
        android:inputType="textPassword"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.loginapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin   = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String user = etUsername.getText().toString().trim();
                String pass = etPassword.getText().toString().trim();
                if (user.isEmpty() || pass.isEmpty()) {
                    Toast.makeText(MainActivity.this,
                        "Please fill all fields", Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this,
                        "Welcome " + user, Toast.LENGTH_SHORT).show();
                }
```

```
            }
        });
    }
}
```

Q4. AutoCompleteTextView – Sixth Semester Subjects

Paste XML in res/layout/activity_main.xml and Java in MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Search Subject:"
        android:textSize="16sp" android:textStyle="bold"/>

    <AutoCompleteTextView
        android:id="@+id/autoCompleteTv"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Type subject name..."
        android:completionThreshold="1"
        android:textSize="16sp"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.autocomplete;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.AutoCompleteTextView;

public class MainActivity extends AppCompatActivity {
```

```
    String[] subjects = {
        "Mobile Application Development",
        "Cloud Computing",
        "Big Data Analytics",
        "Machine Learning",
        "Information Security",
        "Software Project Management",
        "Internet of Things"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        AutoCompleteTextView acTv = findViewById(R.id.autoCompleteTv);

        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
            android.R.layout.simple_list_item_1, subjects);

        acTv.setAdapter(adapter);
    }
}
```

Q5. Toggle Button – Bluetooth ON / OFF

Add permission in AndroidManifest.xml: <uses-permission android:name="android.permission.BLUETOOTH"/>

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tvStatus"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Bluetooth: OFF"
```

```
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <ToggleButton
        android:id="@+id/toggleBluetooth"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Bluetooth ON"
        android:textOff="Bluetooth OFF"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.bluetooth;

import androidx.appcompat.app.AppCompatActivity;
import android.bluetooth.BluetoothAdapter;
import android.os.Bundle;
import android.widget.CompoundButton;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.ToggleButton;

public class MainActivity extends AppCompatActivity {

    ToggleButton toggleBluetooth;
    TextView tvStatus;
    BluetoothAdapter bluetoothAdapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toggleBluetooth = findViewById(R.id.toggleBluetooth);
        tvStatus         = findViewById(R.id.tvStatus);
        bluetoothAdapter = BluetoothAdapter.getDefaultAdapter();

        toggleBluetooth.setOnCheckedChangeListener(
            new CompoundButton.OnCheckedChangeListener() {
            @Override
```

```
            public void onCheckedChanged(CompoundButton btn, boolean isChecked) {
                if (isChecked) {
                    bluetoothAdapter.enable();
                    tvStatus.setText("Bluetooth: ON");
                    Toast.makeText(MainActivity.this, "Bluetooth ON",
                        Toast.LENGTH_SHORT).show();
                } else {
                    bluetoothAdapter.disable();
                    tvStatus.setText("Bluetooth: OFF");
                    Toast.makeText(MainActivity.this, "Bluetooth OFF",
                        Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

Q6. Login Form for Student Registration System

Paste XML in res/layout/activity_main.xml and Java in MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Student Registration Login"
        android:textSize="22sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:layout_marginBottom="24dp"/>

    <EditText
        android:id="@+id/etName"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Full Name"
        android:inputType="textPersonName"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etRoll"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Roll Number"
        android:inputType="number"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register"
        android:backgroundTint="#1565C0"
        android:textColor="#FFFFFF"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.studentlogin;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText etName, etRoll, etEmail, etPassword;
    Button btnRegister;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName     = findViewById(R.id.etName);
        etRoll     = findViewById(R.id.etRoll);
        etEmail    = findViewById(R.id.etEmail);
        etPassword = findViewById(R.id.etPassword);
        btnRegister = findViewById(R.id.btnRegister);

        btnRegister.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                String name = etName.getText().toString().trim();
                String roll = etRoll.getText().toString().trim();
                String email = etEmail.getText().toString().trim();
                String pass = etPassword.getText().toString().trim();

                if (name.isEmpty()||roll.isEmpty()||email.isEmpty()||pass.isEmpty()) {
                    Toast.makeText(MainActivity.this,
                        "All fields required", Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this,
                        "Registered: " + name, Toast.LENGTH_LONG).show();
                }
            }
        });
    }
}
```

Q7. Five CheckBoxes – Toast Selected Items

Paste XML in res/layout/activity_main.xml and Java in MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout width="match parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <CheckBox android:id="@+id/cb1" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Cricket"
android:textSize="18sp"/>

    <CheckBox android:id="@+id/cb2" android:layout width="wrap content"
        android:layout_height="wrap_content" android:text="Football"
android:textSize="18sp"/>

    <CheckBox android:id="@+id/cb3" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Basketball"
android:textSize="18sp"/>

    <CheckBox android:id="@+id/cb4" android:layout width="wrap content"
        android:layout_height="wrap_content" android:text="Tennis"
android:textSize="18sp"/>

    <CheckBox android:id="@+id/cb5" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:text="Badminton"
android:textSize="18sp"/>

    <Button android:id="@+id/btnSubmit" android:layout width="match parent"
        android:layout_height="wrap_content" android:text="Submit"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.checkboxapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;
```

```
public class MainActivity extends AppCompatActivity {

    CheckBox cb1, cb2, cb3, cb4, cb5;
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        cb1 = findViewById(R.id.cb1);  cb2 = findViewById(R.id.cb2);
        cb3 = findViewById(R.id.cb3);  cb4 = findViewById(R.id.cb4);
        cb5 = findViewById(R.id.cb5);
        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                StringBuilder sb = new StringBuilder("Selected: ");
                CheckBox[] boxes = {cb1, cb2, cb3, cb4, cb5};
                boolean any = false;
                for (CheckBox cb : boxes) {
                    if (cb.isChecked()) {
                        sb.append(cb.getText()).append(", ");
                        any = true;
                    }
                }
                String msg = any ? sb.toString() : "Nothing selected";
                Toast.makeText(MainActivity.this, msg, Toast.LENGTH_LONG).show();
            }
        });
    }
}
```

Q8. ListView – Programming Languages

Paste XML in res/layout/activity_main.xml and Java in MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:orientation="vertical">


    <TextView android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="ListView"
        android:textSize="20sp" android:textStyle="bold"
        android:background="#1565C0" android:textColor="#FFFFFF"
        android:padding="12dp"/>


    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>


</LinearLayout>
```

File: MainActivity.java

```
package com.example.listviewapp;


import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.*;


public class MainActivity extends AppCompatActivity {


    String[] languages = {"Android","Java","Php","Hadoop","Sap",
                          "Python","Ajax","C++","Ruby","Rails",".Net"};


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        ListView listView = findViewById(R.id.listView);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
            android.R.layout.simple_list_item_1, languages);
        listView.setAdapter(adapter);


        listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, android.view.View view,
```

```
                                    int position, long id) {
                Toast.makeText(MainActivity.this,
                    "Selected: " + languages[position], Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Q9. CheckBoxes – Pizza / Coffee / Burger with Order Button

Toast shows selected items, individual price and total price on ORDER button click

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">

    <CheckBox android:id="@+id/cbPizza"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Pizza" android:textSize="18sp"/>

    <CheckBox android:id="@+id/cbCoffee"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Coffee" android:textSize="18sp"/>

    <CheckBox android:id="@+id/cbBurger"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Burger" android:textSize="18sp"/>

    <Button android:id="@+id/btnOrder"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="ORDER"
        android:layout_marginTop="30dp"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.orderapp;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    CheckBox cbPizza, cbCoffee, cbBurger;
    Button btnOrder;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        cbPizza  = findViewById(R.id.cbPizza);
        cbCoffee = findViewById(R.id.cbCoffee);
        cbBurger = findViewById(R.id.cbBurger);
        btnOrder = findViewById(R.id.btnOrder);

        btnOrder.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                StringBuilder sb = new StringBuilder("Selected Items:\n");
                int total = 0;
                if (cbPizza.isChecked())  { sb.append("Pizza  80Rs\n");  total += 80;  }
                if (cbCoffee.isChecked()) { sb.append("Coffe  50Rs\n");  total += 50;  }
                if (cbBurger.isChecked()) { sb.append("Burger 120Rs\n"); total += 120; }
                if (total == 0) {
                    Toast.makeText(MainActivity.this, "No item selected",
                        Toast.LENGTH SHORT).show();
                } else {
                    sb.append("Total: ").append(total).append("Rs");
                    Toast.makeText(MainActivity.this, sb.toString(),
                        Toast.LENGTH_LONG).show();
                }
            }
        });
    }
}
```

Q10. TimePicker with Spinner Mode

Paste XML in res/layout/activity_main.xml and Java in MainActivity.java

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:timePickerMode="spinner"/>

    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Time"
        android:layout_marginTop="20dp"/>

    <TextView
        android:id="@+id/tvTime"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="36sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:layout_marginTop="16dp"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.timepicker;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;
```

```
public class MainActivity extends AppCompatActivity {

    TimePicker timePicker;
    Button btnShow;
    TextView tvTime;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        timePicker = findViewById(R.id.timePicker);
        btnShow    = findViewById(R.id.btnShow);
        tvTime     = findViewById(R.id.tvTime);

        timePicker.setIs24HourView(true);   // 24-hour mode

        btnShow.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                int hour   = timePicker.getHour();   // API 23+
                int minute = timePicker.getMinute();
                String time = String.format("%02d:%02d", hour, minute);
                tvTime.setText(time);
            }
        });
    }
}
```

Q11. Two Screens – Factorial using Explicit Intent

Intent Type Used: EXPLICIT INTENT – because we know the target Activity class name.
Create two activities: MainActivity and ResultActivity. Steps below.

File: res/layout/activity_main.xml (Screen 1)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:gravity="center" android:padding="24dp">

    <TextView android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Enter Number:" android:textSize="18sp"/>
```

```
    <EditText android:id="@+id/etNumber"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Number" android:inputType="number"/>

    <Button android:id="@+id/btnFactorial"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="Factorial" android:layout_marginTop="16dp"/>

</LinearLayout>
```

File: MainActivity.java (Screen 1)

```
package com.example.factorial;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText etNumber;
    Button btnFactorial;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etNumber     = findViewById(R.id.etNumber);
        btnFactorial = findViewById(R.id.btnFactorial);

        btnFactorial.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                String numStr = etNumber.getText().toString().trim();
                if (numStr.isEmpty()) {
                    Toast.makeText(MainActivity.this,"Enter a
number",Toast.LENGTH_SHORT).show();
                    return;
                }
                int num = Integer.parseInt(numStr);
```

```
                // Explicit Intent - we name the target Activity directly
                Intent intent = new Intent(MainActivity.this, ResultActivity.class);
                intent.putExtra("NUMBER", num);
                startActivity(intent);
            }
        });
    }
}
```

File: res/layout/activity_result.xml (Screen 2)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:gravity="center" android:orientation="vertical" android:padding="24dp">

    <TextView android:id="@+id/tvResult"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:textSize="24sp" android:textStyle="bold" android:textColor="#1565C0"/>

</LinearLayout>
```

File: ResultActivity.java (Screen 2)

```
package com.example.factorial;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class ResultActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_result);

        int num = getIntent().getIntExtra("NUMBER", 0);
        long fact = factorial(num);

        TextView tvResult = findViewById(R.id.tvResult);
        tvResult.setText("Factorial of " + num + " = " + fact);
    }

    private long factorial(int n) {
```

```
        if (n <= 1) return 1;
        return n * factorial(n - 1);
    }
}
```

IMPORTANT: Register ResultActivity in AndroidManifest.xml inside <application> tag:

```
<activity android:name=".ResultActivity"></activity>
```

Q12. Start Wi-Fi using Service

Add these permissions in AndroidManifest.xml:

```
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE"/>
```

File: WifiService.java (new Java class – right-click package > New > Java Class)

```
package com.example.wifiservice;

import android.app.Service;
import android.content.Context;
import android.content.Intent;
import android.net.wifi.WifiManager;
import android.os.IBinder;
import android.widget.Toast;

public class WifiService extends Service {

    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        WifiManager wifiManager = (WifiManager)
            getApplicationContext().getSystemService(Context.WIFI_SERVICE);

        if (!wifiManager.isWifiEnabled()) {
            wifiManager.setWifiEnabled(true);   // deprecated in API 29+
            Toast.makeText(this, "Wi-Fi Enabled", Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "Wi-Fi already ON", Toast.LENGTH_SHORT).show();
        }
        return START_STICKY;
    }

    @Override public IBinder onBind(Intent intent) { return null; }
}
```

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout width="match parent" android:layout height="match parent"
    android:gravity="center" android:orientation="vertical">

    <Button android:id="@+id/btnStartWifi"
        android:layout width="wrap content" android:layout height="wrap content"
        android:text="Start Wi-Fi Service"/>

</LinearLayout>
```

File: MainActivity.java

```
package com.example.wifiservice;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.btnStartWifi);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                Intent serviceIntent = new Intent(MainActivity.this, WifiService.class);
                startService(serviceIntent);
            }
        });
    }
}
```

Register the service in AndroidManifest.xml inside <application> tag:

```
<service android:name=".WifiService"/>
```

Q13. Insert Data in SQLite Database using AsyncTask

Create three files: DatabaseHelper.java, InsertTask.java, MainActivity.java + XML layout.

File: DatabaseHelper.java (new Java class in same package)

```
package com.example.sqliteasync;


import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;


public class DatabaseHelper extends SQLiteOpenHelper {


    static final String DB_NAME    = "Students.db";
    static final int    DB_VERSION = 1;
    static final String TABLE      = "students";


    public DatabaseHelper(Context ctx) {
        super(ctx, DB_NAME, null, DB_VERSION);
    }


    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE " + TABLE +
            "(_id INTEGER PRIMARY KEY AUTOINCREMENT," +
            " name TEXT, roll TEXT, marks TEXT)");
    }


    @Override
    public void onUpgrade(SQLiteDatabase db, int oldV, int newV) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE);
        onCreate(db);
    }
}
```

File: InsertTask.java (new Java class – AsyncTask)

```
package com.example.sqliteasync;


import android.content.ContentValues;
import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.os.AsyncTask;
import android.widget.Toast;


public class InsertTask extends AsyncTask<String, Void, Boolean> {
```

```
    Context context;
    DatabaseHelper dbHelper;

    public InsertTask(Context ctx) {
        this.context  = ctx;
        this.dbHelper = new DatabaseHelper(ctx);
    }

    @Override
    protected Boolean doInBackground(String... params) {
        // params[0]=name, params[1]=roll, params[2]=marks
        SQLiteDatabase db = dbHelper.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("name",  params[0]);
        cv.put("roll",  params[1]);
        cv.put("marks", params[2]);
        long result = db.insert(DatabaseHelper.TABLE, null, cv);
        db.close();
        return result != -1;
    }

    @Override
    protected void onPostExecute(Boolean success) {
        if (success)
            Toast.makeText(context, "Data Inserted Successfully!",
Toast.LENGTH_SHORT).show();
        else
            Toast.makeText(context, "Insertion Failed!", Toast.LENGTH_SHORT).show();
    }
}
```

File: res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="24dp" android:gravity="center">

    <EditText android:id="@+id/etName"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Student Name" android:inputType="textPersonName"/>

    <EditText android:id="@+id/etRoll"
```

```
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Roll Number" android:inputType="number"
android:layout_marginTop="8dp"/>


    <EditText android:id="@+id/etMarks"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Marks" android:inputType="number" android:layout_marginTop="8dp"/>


    <Button android:id="@+id/btnInsert"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="Insert Data" android:layout_marginTop="20dp"/>


</LinearLayout>
```

File: MainActivity.java

```
package com.example.sqliteasync;


import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;


public class MainActivity extends AppCompatActivity {


    EditText etName, etRoll, etMarks;
    Button btnInsert;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        etName   = findViewById(R.id.etName);
        etRoll   = findViewById(R.id.etRoll);
        etMarks  = findViewById(R.id.etMarks);
        btnInsert = findViewById(R.id.btnInsert);


        btnInsert.setOnClickListener(new View.OnClickListener() {
            @Override public void onClick(View v) {
                String name  = etName.getText().toString().trim();
                String roll  = etRoll.getText().toString().trim();
                String marks = etMarks.getText().toString().trim();
```

```
                if (name.isEmpty() || roll.isEmpty() || marks.isEmpty()) {
                    Toast.makeText(MainActivity.this,
                        "All fields required", Toast.LENGTH_SHORT).show();
                    return;
                }
                // Execute AsyncTask
                new InsertTask(MainActivity.this).execute(name, roll, marks);
            }
        });
    }
}
```

— End of MAD Practical Solutions —